



## On transitive sets of derangements in primitive groups

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**Abstract.** We construct a primitive permutation action of the Steinberg triality group  ${}^3D_4(2)$  of degree 4,064,256 and show that there are distinct points  $\alpha, \beta$  such that there is no derangement  $g \in {}^3D_4(2)$  with  $\alpha^g = \beta$ . This answers a question by John G. Thompson (Problem 8.75 in the Kurovka Notebook) in the negative.

**Mathematics Subject Classification.** 20B15.

**Keywords.** Primitive permutation groups, Derangements, Steinberg triality group.

**1. Introduction.** Beginning with the 8th edition in 1982, the Kurovka Notebook [9], which collects unsolved problems in group theory, has included the following question—attributed to John G. Thompson and designated “A known problem”—as Problem 8.75:

**Question 1.1.** *Suppose  $G$  is a finite primitive permutation group on  $\Omega$ , and  $\alpha, \beta$  are distinct points of  $\Omega$ . Does there exist an element  $g \in G$  such that  $\alpha^g = \beta$  and  $g$  fixes no point of  $\Omega$ ?*

A potential interest of this is that if the answer were affirmative with a proof not using character theory, then Frobenius’ theorem that the kernel of a finite Frobenius group is a subgroup would admit a proof devoid of character theory. Such a proof still does not exist, despite some progress like [4] and [14]. See the argument in [8, Assertion 2] (which relies on [4]) how Frobenius’ theorem would follow from a positive answer of Question 1.1.

We refer to the survey [2] for more open problems (like this one in [2, Remark 1.1.3(iii)]) and results about derangements in permutation groups.

For most Aschbacher–ONan–Scott types of primitive groups the answer to Question 1.1 is trivially yes in view of the presence of regular subgroups. As we considered it unlikely to find a counterexample for primitive groups of

product type (Class III in [11]), our search focused on the almost simple action of groups and eventually produced a counterexample:

**Theorem 1.2.** *The Steinberg triality group  $G = {}^3D_4(2)$  of order  $211,341,312 = 2^{12} \cdot 3^4 \cdot 7^2 \cdot 13$  admits a primitive permutation action on a set  $\Omega$  of size  $4,064,256$  with the following property: There exist distinct elements  $\alpha, \beta \in \Omega$  such that every  $g \in G$  with  $\alpha^g = \beta$  has at least one fixed point in  $\Omega$ .*

This counterexample is the only one we know of. Let  $\text{soc}(G)$  be the socle of an almost simple group  $G$ . There are 584 nonabelian simple groups  $S$  of order less than or equal to 20,158,709,760. A brute force computation using Magma [1], where  $S = \text{soc}(G)$  is one of these simple groups, resulted in the only example from Theorem 1.2. (The cases  $S = \text{PSL}(2, q)$  were only checked for prime powers  $q \leq 1789$ . But this case can be ruled out anyway, see Remark (g).) We also used the data base of primitive permutation groups in Magma [1] to check that there is no counterexample of degree less than 8192.

In contrast to the demanding search for a counterexample, the proof of Theorem 1.2 will require little computation. We will work with  ${}^3D_4(2)$  as a subgroup of  $\text{GL}_8(\mathbb{F}_8)$ . Except for some general group theoretic arguments and the known fact that  ${}^3D_4(2)$  has a maximal subgroup isomorphic to  $C_{13} \rtimes C_4$ , the verification of Theorem 1.2 is reduced to check some identities for specific matrices from  $\text{GL}_8(\mathbb{F}_8)$ .

**2. Proof of Theorem 1.2.** The proof will be based on the following lemma.

**Lemma 2.1.** The Steinberg triality group  $G = {}^3D_4(2)$  contains elements  $s, x$ , and  $y$  with the following properties:

- (1)  $x, y$ , and  $yx$  are conjugate in  $G$ ,
- (2)  $x^y = x^{-1}$ ,
- (3)  $x^2 = y^2$ ,
- (4)  $s$  has order 13,
- (5)  $s^y = s^8$ ,
- (6)  $x \notin \langle s, y \rangle$ ,
- (7)  $(hx)^{13} = 1$  for every  $h \in \langle s, y \rangle \setminus \langle y \rangle$ .

Before proving this lemma, we show how Theorem 1.2 follows from it. By (4) and (5),  $y$  acts as an automorphism of order 4 on the Sylow 13-subgroup  $S = \langle s \rangle$ . According to [10],  $G$  has a maximal subgroup isomorphic to  $C_{13} \rtimes C_4$ . Thus  $y$  has order 4, and  $G$  acts primitively on the  $|G|/52 = 4,064,256$  right cosets of  $H = \langle s, y \rangle = \langle s \rangle \rtimes \langle y \rangle$  in  $G$ .

Note that  $H \neq Hx$  by (6). We claim that every  $g \in Hx$  fixes a right coset of  $H$ . Write  $g = hx$  for some  $h \in H$ . We need to show that  $hx$  is contained in some conjugate  $H^z$  of  $H$ , for then  $Hzg = Hzhx = Hz$ .

If  $h \in \langle s, y \rangle \setminus \langle y \rangle$ , then  $hx$  lies in a conjugate of  $S < H$  by (7).

It remains to look at the case  $h \in \langle y \rangle$ , that is,  $hx \in \{x, yx, y^2x, y^{-1}x\}$ . By (1),  $x$  and  $yx$  are conjugate to  $y \in H$ . From (2), we get  $y^{-1}x = x^{-1}y^{-1} = (yx)^{-1}$ , so  $y^{-1}x$  is conjugate to  $y^{-1} \in H$  by (1). Finally,  $y^2x = x^3$  by (3), so  $y^2x$  is conjugate to  $y^3 \in H$  by (1).

**2.1. Proof of Lemma 2.1.** We work with a representation of  $G = {}^3D_4(2)$  as a subgroup of  $\text{GL}_8(\mathbb{F}_8)$  as given in [7, Section 4.2]: Pick  $\omega \in \mathbb{F}_8$  with  $\omega^3 + \omega + 1 = 0$ . Note that  $\omega$  has multiplicative order 7. Let  $E_{i,j}$  be the  $8 \times 8$  matrix over  $\mathbb{F}_8$  with 1 in the  $i$ th row and  $j$ th column and 0 elsewhere. Furthermore, let  $E$  be the  $8 \times 8$  identity matrix. Using the notation from [7], we set

$$\begin{aligned} x_R(1) &= E + E_{1,2} + E_{3,4} + E_{3,5} + E_{3,6} + E_{4,6} + E_{5,6} + E_{7,8}, \\ n &= E_{1,3} + E_{2,1} + E_{3,7} + E_{4,5} + E_{5,4} + E_{6,2} + E_{7,8} + E_{8,6}, \\ h_R(\omega) &= \text{diagonal matrix with diagonal } (\omega^4, \omega^3, \omega^3, \omega, \omega, \omega^6, \omega^4, \omega^4, \omega^3). \end{aligned}$$

Then  $G = \langle a, b \rangle$ , where  $a = x_R(1) \cdot n$  and  $b = h_R(\omega)$ .

Essentially the same generators  $a$  and  $b$  are given in [15, Abstract group 211341312.a], the only difference is that the fourth and fifth rows and columns are switched there.

With these matrices  $a$  and  $b$ , set

$$\begin{aligned} x &= b \cdot (a^2 \cdot b^2 \cdot a)^2 \cdot b^{-2} \cdot a \cdot b^{-2}, \\ t &= a^{-3} \cdot b \cdot (b \cdot a)^2 \cdot b^{-2} \cdot a^2 \cdot b, \\ y &= x^t, \\ s &= a \cdot a^b, \\ d &= x \cdot s \cdot y \cdot s \cdot a \cdot y \cdot s^2 \cdot t. \end{aligned}$$

With these settings, the following SageMath [16] code verifies all the claims instantly (in less than a second). The comments indicate the respective items from Lemma 2.1. If one wants to confirm that the matrices  $a$  and  $b$  are correctly copied (and correctly given in [7]), one may uncomment the last line. Note that except for the last line, the code merely does simple matrix operations which in principle could be done by hand.

```
F.<w> = GF(8, modulus=[1, 1, 0, 1])
e = matrix.identity(F, 8)
xr1_ij = [(1, 2), (3, 4), (3, 5), (3, 6), (4, 6), (5, 6), (7, 8)]
n_ij = [(1, 3), (2, 1), (3, 7), (4, 5), (5, 4), (6, 2), (7, 8), (8, 6)]
xr1 = e + matrix(F, 8, {(i-1, j-1): 1 for i, j in xr1_ij})
n = matrix(F, 8, {(i-1, j-1): 1 for i, j in n_ij})
a = xr1 * n
b = matrix.diagonal(F, [w^z for z in [4, 3, 3, 1, 6, 4, 4, 3]])
x = b * (a^2 * b^2 * a)^2 * b^-2 * a * b^-2
t = a^-3 * b * (b * a)^2 * b^-2 * a^2 * b
y = t^-1 * x * t                                     # one part of (1)
s = a * b^-1 * a * b
d = x * s * y * s * a * y * s^2 * t
S = [s^i for i in range(13)]
Y = [y^i for i in range(4)]
H = [u * v for u in S for v in Y]
assert d^-1 * x * d == y * x                         # other part of (1)
assert y^-1 * x * y == x^-1                         # (2)
assert x^2 == y^2                                    # (3)
assert s != e and s^13 == e                          # (4)
assert y^-1 * s * y == s^8                          # (5)
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assert x not in H # (6)
assert all(h in Y or (h * x)^13 == e for h in H) # (7)
print('All claims verified')
# Uncomment to optionally check that a and b indeed generate 3D(4,2):
# assert MatrixGroup([a, b]).structure_description() == '3D(4,2)'

```

This SageMath code is provided at [12], where it can be downloaded and run online.

### 3. Remarks.

- (a) If we replace  $G$  by its automorphism group  ${}^3D_4(2).3$ , then the primitive action of degree 4,064,256 has a transitive set of derangements, so it is not a counterexample to Question 1.1.
- (b) Let  $H$  be a subgroup of a finite group  $G$ . Then the inclusion  $Hx \subseteq \cup_{g \in G} H^g$  holds for  $x \in G$  if and only if  $HxH \subseteq \cup_{g \in G} H^g$ . For  $G$  and a point stabilizer  $H$  as in Theorem 1.2, there are precisely 78,366 double cosets  $HxH$  different from  $H$ , and it turns out that exactly one of these double cosets accounts for the counterexample!
- (c) In hindsight, this kind of uniqueness of the counterexample helped to derive some of the structural properties in Lemma 2.1. For instance, if  $HxH$  gives a counterexample, then so does  $Hx^{-1}H$ , thus  $HxH = Hx^{-1}H$  by uniqueness. So  $x^{-1} = h_1 x h_2$  for  $h_i \in H$ . This gives  $(xh_1)^{-1} = (xh_1)(h_1^{-1}h_2)$ . So upon replacing  $x$  with  $xh_1$ , we may assume  $x^2 \in H$ . Now  $x$  has order  $\neq 13$ , for otherwise  $x = (x^2)^7 \in H$ . Thus  $x$  has order 2 or 4. One can rule out the former case. Again, let  $S$  be the Sylow 13-subgroup of  $H$ . The normalizer of  $S$  in the automorphism group  ${}^3D_4(2).3$  of  $G$  has the form  $S \rtimes \langle \sigma \rangle$  with  $\sigma$  of order 12. Clearly,  $\sigma$  normalizes  $H$ , and we may assume that  $\sigma^6 = x^2$ . Here  $\sigma^3$  plays the role of  $y$  in Lemma 2.1. From  $Hx^\sigma H = HxH$ , we get further properties from Lemma 2.1.
- (d) I have tried to extend the counterexample to the bigger twisted triality groups  $G = {}^3D_4(q)$  for  $q = 3$  and  $q = 4$  and the maximal subgroup  $H = C_{q^4 - q^2 + 1} \rtimes C_4$ . Here,  $G$  is far too large for it to be feasible to compute representatives  $x$  of the double cosets of  $H$  in  $G$ . I was only able to verify that the elements  $x$  of order 4 with  $x^2 \in H$  do not produce counterexamples (unlike in the case  $q = 2$ ).
- (e) Initially, I had tried to explore a vast generalization of Question 1.1. The primitivity of a group can not be read off from the permutation character, see the negative answer [5] by Guralnick and Saxl to a question of Wielandt. However, by a result of Higman (see [6] or [3, Theorem 3.2A]), primitivity is encoded in the orbital graphs: The group  $G$  is primitive if and only if each non-diagonal orbital graph is connected. Now there was some experimental evidence that the following generalization of Question 1.1 could hold: Let  $G$  act transitively on the finite set  $\Omega$ . Pick distinct  $\alpha, \beta$  in  $\Omega$ . If the orbital graph which contains the edge  $(\alpha, \beta)$  is connected, then there is a derangement  $g \in G$  with  $\alpha^g = \beta$ . The smallest counterexample which I found is  $G = \text{PSL}_3(4)$  acting on 4032 points (so the point stabilizer has order 5).

- (f) I have some evidence (and partial results) that the following could hold: Let  $G$  act transitively on the finite set  $\Omega$ , and pick distinct  $\alpha, \beta \in \Omega$ . Then there is an element  $g \in G$  with  $\alpha^g = \beta$  whose number of fixed points is different from 1. This is now Problem 21.99 in the 21st edition (2026) of the Kourovka Notebook [9].  
Applied to a finite Frobenius permutation group acting on a finite set  $\Omega$ , this would mean that for any distinct  $\alpha, \beta \in \Omega$ , there is a unique derangement  $g \in G$  with  $\alpha^g = \beta$ . But then Frobenius' theorem follows immediately from a simple counting argument (see e.g. [8, Assertion 1]).
- (g) It would still be interesting to decide for which primitive groups Question 1.1 has a positive answer. With some effort, one can show that this is the case if  $G$  is almost simple with socle  $\text{soc}(G) = \text{PSL}_2(\mathbb{F}_q)$ . See [13].

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